PERCEPTA

Perceptual pattern image generator

RGB geometry • optical reconstruction • animated density

Technical Manual

Theory, algorithms, controls and practical use

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1 Purpose, scope and operating workflow

1.1 What Percepta produces

Percepta is a desktop application for transforming an ordinary raster image into a structured RGB pattern. The output is deliberately dual in character. At close range, the observer sees a geometric construction: coloured stripes, dots or a spiral. At a larger viewing distance, or after reducing the image on screen, the same construction is spatially averaged and the source image becomes legible again.

The software therefore does not aim to imitate the source at every pixel. Instead, it preserves local red, green and blue contributions through the *coverage* of three independently rendered structures. In a bright red region, the red structure occupies a larger fraction of the local pattern cell. In a neutral bright region, red, green and blue structures overlap strongly and the area tends towards white. In a dark region, little geometry is drawn and the background remains visible.

The intended uses include:

- creating optical-illusion portraits and graphic artwork;
- producing animated transitions between obvious geometry and image reconstruction;
- exploring spatial vision, colour mixing and halftoning principles;
- preparing high-resolution raster or vector compositions for screen and print.

Principle

Percepta converts **intensity into geometry**. It does not merely recolour pixels or overlay a fixed screen. The source controls local stripe width, dot radius or spiral thickness separately in the red, green and blue channels.

1.2 Document scope

This manual provides a complete operational and mathematical description of the current application. It explains the user interface, preprocessing, rendering families, parameter semantics, adaptive defaults, density animation, export and the perceptual basis of the illusion.

The equations describe the rendering model at the level needed to understand and reproduce its behaviour. A concrete implementation may use equivalent discrete formulations, cached intermediate images, vector paths or antialiased raster primitives. Consequently, two valid implementations can differ by subpixel edge placement while obeying the same model.

1.3 Main interface

Figure 1 reconstructs the application layout from the supplied interface captures. The exact dimensions adapt to the display, but the functional organisation remains the same.

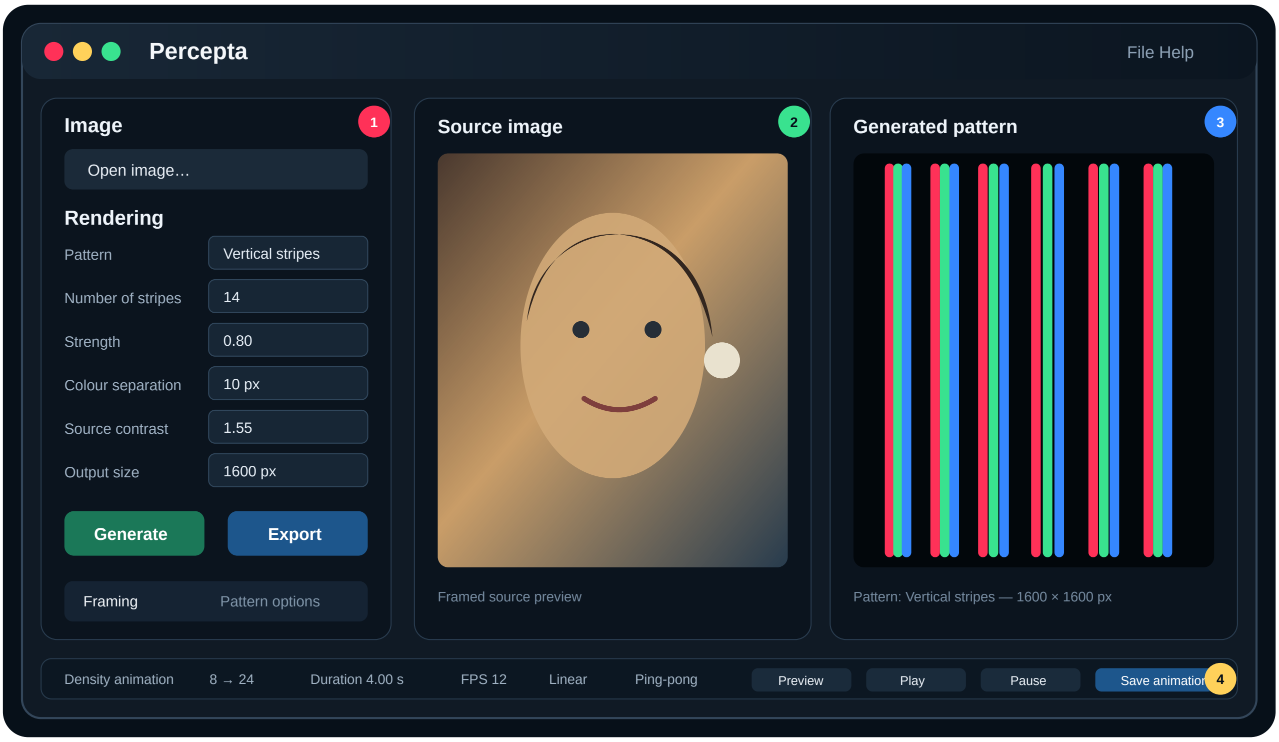

Figure 1: Percepta interface: (1) input and rendering controls, (2) framed source preview, (3) generated pattern preview, and (4) density animation controls.

1.3.1 Input and rendering panel

The left panel contains the global workflow:

1. **Open image...** loads the source.
2. **Pattern** selects one of the six renderers.
3. The density field changes its label and unit according to the renderer.
4. **Strength**, **Colour separation**, **Source contrast** and **Output size** set common behaviour.
5. **Generate** refreshes the output.

6. **Export** writes the current result to a supported format.

The two lower tabs separate geometric framing from options that apply only to the chosen pattern. This avoids displaying irrelevant controls and makes it clear which settings are global and which are renderer-specific.

1.3.2 Source and generated previews

The source preview is not necessarily the original file as loaded. It represents the result of crop, zoom, pan and rotation. This is the exact composition passed to the encoder. The generated preview shows the geometric output and reports the active pattern and output dimensions.

A useful evaluation habit is to inspect the generated result in three ways:

- at full size, to judge edge quality and the visual character of the geometry;
- at an intermediate zoom, to see whether pattern and image coexist;
- as a small thumbnail, to judge reconstruction.

1.4 Recommended operating sequence

The following sequence minimises unnecessary recalculation and makes parameter interactions easier to understand.

Step 1. Choose a suitable source. Begin with an image having a clear subject, useful tonal range and limited clutter.
Step 2. Set the output aspect ratio. Select the crop before fine positioning.
Step 3. Frame the subject. Use zoom, pan and rotation until the source preview is satisfactory.
Step 4. Choose the pattern family. Select a line, halftone or spiral structure according to the intended visual character.
Step 5. Generate at reference settings. At 1600 px, use the pattern defaults in Table 1 before making large changes.
Step 6. Adjust density first. Density determines the fundamental spatial scale and has the largest effect on recognisability.
Step 7. Adjust Strength and contrast. Use them to recover tonal separation without clipping.
Step 8. Add colour separation carefully. Coloured fringes are attractive, but excessive displacement breaks local RGB averaging.
Step 9. Test at distance. Zoom out or physically step back.
Step 10. Increase output size for export. Adaptive defaults preserve approximate apparent density, but the final result should still be checked.

Practical note

For a first portrait, use Vertical stripes at 14 stripes, Strength 0.80, Colour separation 10 px, Source contrast 1.55 and Output size 1600 px. Then compare the same framing with Halftone at 12 px and Spiral stripes at 19 px.

1.5 Pattern overview

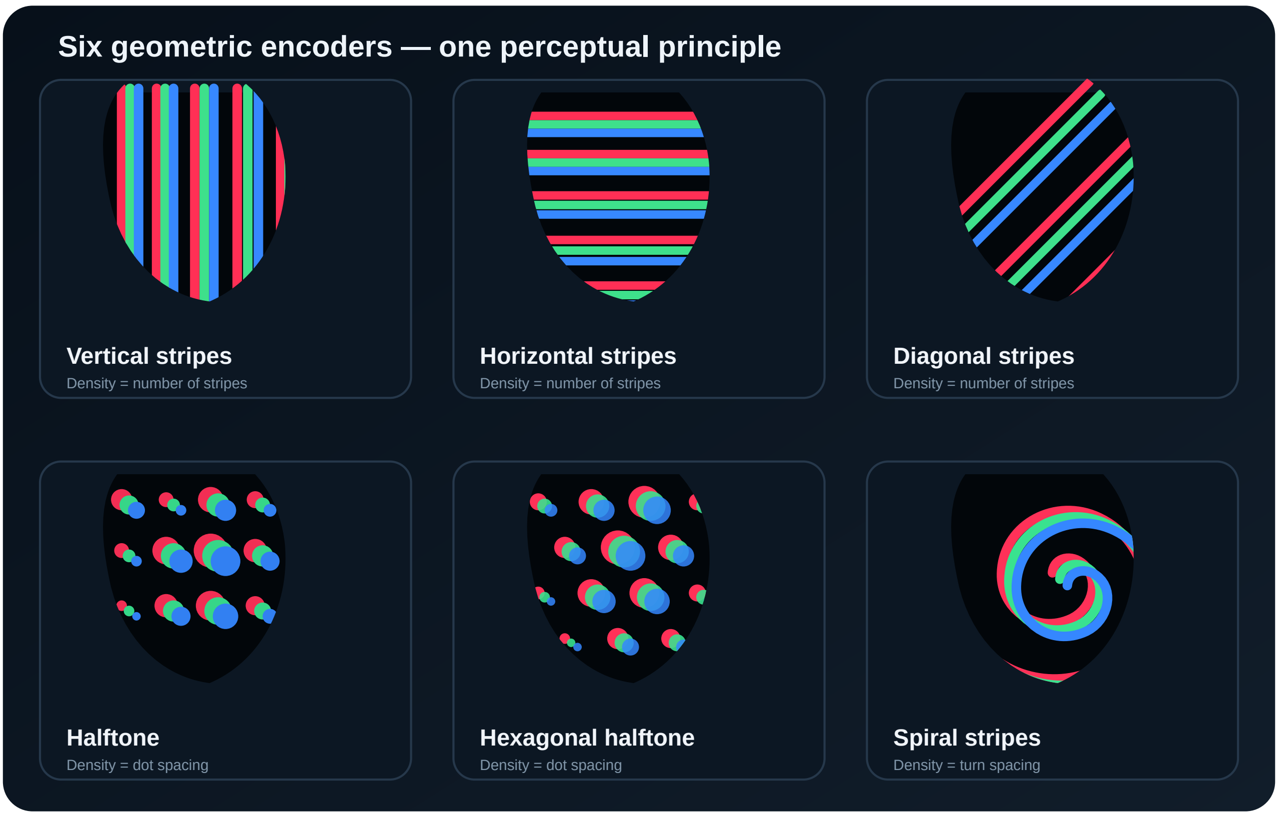

Figure 2: The six renderer families. The illustrations reproduce the visual logic of the supplied Percepta examples rather than a specific source photograph.

The line renderers use one-dimensional carriers whose local visible width changes with image content. The two halftone renderers sample on two-dimensional lattices and encode intensity as dot size. The spiral renderer follows a single continuous polar path and varies its local thickness.

These families share the same RGB decomposition and additive composition, so the controls remain conceptually consistent even though their density units differ.

2 Image preparation and mathematical encoding model

2.1 Coordinate system and framed source

Let the loaded source image be defined on pixel coordinates (u, v). Percepta first constructs an output canvas of width W and height H. A point (x, y) on that canvas is mapped back to the source through crop, zoom, translation and rotation.

Using homogeneous coordinates, the combined inverse mapping can be written as

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{T}_{\text{source}} \mathbf{R}(-\varphi) \mathbf{S}\left(\frac{1}{z}\right) \mathbf{T}_{\text{pan}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}, \tag{1}$$

where z is the zoom factor, φ is the rotation angle, and the translation matrices include crop centring and the user-defined horizontal and vertical pan. The source value at non-integer (u, v) is obtained by interpolation. Bilinear interpolation is sufficient for interactive work; higher-order resampling may be used for final export.

The framed RGB image is denoted

$$\mathbf{C}(x, y) = \begin{bmatrix} R(x, y) \\ G(x, y) \\ B(x, y) \end{bmatrix}, \qquad 0 \le R, G, B \le 1. \tag{2}$$

Values are normalised to the interval $[0, 1]$ before rendering.

2.2 Contrast preprocessing

The **Source contrast** parameter changes the distance of each channel value from a mid-grey pivot. A convenient operational form is

$$C'_q(x, y) = \operatorname{clamp}\left[\frac{1}{2} + \gamma_c \left(C_q(x, y) - \frac{1}{2}\right), 0, 1\right], \qquad q \in \{R, G, B\}, \tag{3}$$

where $\gamma_c = 1$ leaves contrast unchanged, $\gamma_c > 1$ increases contrast and $0 < \gamma_c < 1$ compresses it. The function

$$\operatorname{clamp}(a, 0, 1) = \begin{cases} 0, & a < 0, \\ a, & 0 \le a \le 1, \\ 1, & a > 1 \end{cases} \tag{4}$$

prevents invalid channel values. An implementation can use an equivalent image-library contrast operator, but Equation (3) captures the intended behaviour.

Contrast is applied before geometric encoding. It therefore changes the distribution of widths or radii, not merely the appearance of a completed result. High values expand the number of locations that map to minimum or maximum geometry and can remove subtle tonal information.

2.3 Channel-specific geometric coverage

For every channel q, Percepta creates a geometric mask

$$M_q(x, y) \in [0, 1]. \tag{5}$$

A value of zero means that channel contributes no light at (x, y); a value of one means full channel coverage. Antialiased edges take intermediate values.

The source channel is converted into a modulation amplitude by

$$a_q(x, y) = \operatorname{sat}\left(a_{\min} + s\, C'_q(x, y)\right), \tag{6}$$

where s is related to **Strength**, $a_{\min}$ is the renderer's minimum visible geometry and

$$\operatorname{sat}(a) = \operatorname{clamp}(a, 0, 1). \tag{7}$$

In practice, the geometry may use a nonlinear response

$$a_q(x, y) = \operatorname{sat}\left(a_{\min} + s\, [C'_q(x, y)]^{\eta}\right), \tag{8}$$

where η controls how quickly small channel values become visible. The current controls expose Strength rather than η directly; the equation is useful when comparing renderer implementations or future calibration modes.

2.4 RGB displacement

Colour separation is produced by evaluating or placing the channel structures at slightly different positions. Let the separation magnitude in output pixels be d. A symmetric three-channel arrangement can be represented by displacement vectors

$$\boldsymbol{\delta}_R = d \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \qquad \boldsymbol{\delta}_G = d \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \boldsymbol{\delta}_B = d \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \tag{9}$$

or by the same vectors rotated into the local normal direction of a stripe or spiral. The renderer then uses

$$\widetilde{M}_q(x, y) = M_q(x - \delta_{q,x}, y - \delta_{q,y}) \,. \tag{10}$$

The exact orientation can be pattern-dependent. What matters perceptually is that channel supports no longer coincide perfectly. Where two channels overlap, additive secondary colours appear: red plus green gives yellow, green plus blue gives cyan, and red plus blue gives magenta.

2.5 Additive composition

The final RGB output is the vector image

$$\mathbf{I}_{\text{out}}(x, y) = \begin{bmatrix} \widetilde{M}_R(x, y) \\ \widetilde{M}_G(x, y) \\ \widetilde{M}_B(x, y) \end{bmatrix} . \tag{11}$$

On a black background, each mask directly drives the corresponding display primary. If an opacity or background term is introduced, a more general compositing expression is

$$\mathbf{I}_{\text{out}} = \boldsymbol{\alpha} \odot \mathbf{M} + (\mathbf{1} - \boldsymbol{\alpha}) \odot \mathbf{I}_{\text{bg}}, \tag{12}$$

where $\odot$ denotes element-wise multiplication. The standard Percepta appearance uses a dark background because it maximises the visual distinction between individual RGB elements while preserving additive mixing.

2.6 Area-average preservation

Consider a local pattern cell Ω whose dimensions are small compared with the scale of source variation. The mean output colour is

$$\bar{\mathbf{I}}_\Omega = \frac{1}{|\Omega|} \iint_\Omega \mathbf{I}_{\text{out}}(x, y) \, \mathrm{d}x \, \mathrm{d}y. \tag{13}$$

A well-calibrated renderer chooses stripe widths or dot areas so that

$$\bar{\mathbf{I}}_\Omega \approx \begin{bmatrix} \overline{R}_\Omega \\ \overline{G}_\Omega \\ \overline{B}_\Omega \end{bmatrix} . \tag{14}$$

This is the central conservation principle. The pattern can differ radically from the source at individual pixels while preserving average channel energy over a neighbourhood.

Limitation

Exact equality in Equation (14) is not guaranteed everywhere. Minimum widths, clipping, antialiasing, finite spacing, RGB displacement and source variation within a cell all introduce error. The renderer seeks a convincing perceptual approximation rather than lossless coding.

2.7 Sampling and antialiasing

Every ideal geometric mask contains sharp boundaries. Raster output samples those boundaries on a finite pixel grid. Without antialiasing, an ideal binary mask

$$M_q^{(0)}(x, y) \in \{0, 1\} \tag{15}$$

would produce stair-stepping and temporal flicker. A smoothed mask can be modelled as

$$M_q(x, y) = \left(h_{\text{aa}} * M_q^{(0)} \right)(x, y), \tag{16}$$

where h_{aa} is a compact reconstruction kernel and $*$ denotes convolution. Supersampling followed by area downsampling is an equivalent practical method.

Smoothing should remove pixel-grid artefacts without blurring neighbouring RGB structures together. This trade-off is especially important for thin diagonal lines and dense spiral turns.

3 Rendering algorithms

3.1 Common carrier formulation

A renderer defines a carrier coordinate $\xi(x, y)$ and a local period P. The fractional phase is

$$\phi(x, y) = \operatorname{frac}\left(\frac{\xi(x, y)}{P} \right), \qquad 0 \le \phi < 1, \tag{17}$$

where $\operatorname{frac}(a) = a - \lfloor a \rfloor$. A centred stripe with normalised duty cycle a_q is present when the periodic distance from the carrier centre is smaller than half that duty cycle:

$$M_q^{(0)}(x, y) = \begin{cases} 1, & \left| \phi_q(x, y) - \frac{1}{2} \right| \le \frac{a_q(x,y)}{2}, \\ 0, & \text{otherwise.} \end{cases} \tag{18}$$

The channel phase ϕ_q includes Colour separation. Equation (18) is a convenient mathematical description; an implementation can instead draw explicit variable-width polygons or stroked paths.

3.2 Vertical stripes

For N stripes across an output of width W, the nominal horizontal period is

$$P_x = \frac{W}{N}. \tag{19}$$

The carrier coordinate is simply

$$\xi_{\mathrm{V}}(x, y) = x. \tag{20}$$

Local source values are sampled at or around the stripe position. The width of channel q can be expressed as

$$w_q(x, y) = P_x\, a_q(x, y), \qquad 0 \le w_q \le P_x. \tag{21}$$

The strongest variation is normally evaluated along the stripe direction so that the width can change with y. Smooth interpolation between successive samples avoids jagged lateral edges.

Advantages. Vertical stripes are visually simple, computationally efficient and immediately recognisable as a deliberate construction. They often preserve faces well because eyes, mouth and broad horizontal tonal changes intersect many carriers.

Limitations. Very narrow vertical features can align with the gaps between carriers. Increasing N reduces this risk but makes the close-range pattern less bold.

3.3 Horizontal stripes

For N horizontal stripes across height H,

$$P_y = \frac{H}{N}, \qquad \xi_{\mathrm{H}}(x, y) = y. \tag{22}$$

The same modulation is applied with x and y exchanged. Width changes along x, allowing a horizontal carrier to widen and narrow across the image.

Horizontal lines are well suited to landscapes, eye lines and broad lateral structures. They can, however, interact strongly with existing horizons or scan-line textures. Rotation in the Framing tab may be used intentionally to avoid or exploit such alignment.

3.4 Diagonal stripes

Let α be the stripe-normal angle measured from the horizontal image axis. The rotated carrier coordinate is

$$\xi_{\mathrm{D}}(x, y) = x \cos\alpha + y \sin\alpha. \tag{23}$$

The length of the output projected onto that normal is approximately

$$L_\alpha = W|\cos\alpha| + H|\sin\alpha|, \tag{24}$$

so a natural period for N diagonal stripes is

$$P_\alpha = \frac{L_\alpha}{N}. \tag{25}$$

The coordinate tangent to the stripe is

$$\tau_{\mathrm{D}}(x, y) = -x \sin\alpha + y \cos\alpha. \tag{26}$$

Sampling and smoothing are performed along τ_{D}. RGB separation is most coherent when applied along the local normal $(\cos\alpha, \sin\alpha)$.

Diagonal carriers reduce axis alignment with conventional image edges and create a dynamic composition. Because they cross a square canvas over a longer projected distance, the default count is slightly higher than for vertical or horizontal stripes.

3.5 Square-grid halftone

A square halftone uses sample centres

$$\mathbf{p}_{ij} = \begin{bmatrix} x_0 + iP \\ y_0 + jP \end{bmatrix}, \qquad i, j \in \mathbb{Z}, \tag{27}$$

where P is the **Dot spacing**. At each centre, the renderer obtains a representative channel value $c_{q,ij}$, usually by local sampling or averaging. The maximum non-overlapping radius is $P/2$. A linear radius rule is

$$r_{q,ij} = r_{\min} + \left(\frac{P}{2} - r_{\min}\right) s\,[c_{q,ij}]^\eta, \tag{28}$$

followed by clipping to the permitted interval.

However, perceived brightness is more closely related to dot *area* than radius. To make area proportional to channel intensity, use

$$\pi r_{q,ij}^2 = c_{q,ij}\, A_{\max}, \tag{29}$$

which gives

$$r_{q,ij} = r_{\max}\sqrt{c_{q,ij}}. \tag{30}$$

Strength and minimum size can then be incorporated as

$$r_{q,ij} = r_{\min} + (r_{\max} - r_{\min})\sqrt{\mathrm{sat}(s\,c_{q,ij})}. \tag{31}$$

The ideal circular mask is

$$M^{(0)}_{q,ij}(x,y) = \begin{cases} 1, & \|\mathbf{x} - (\mathbf{p}_{ij} + \boldsymbol{\delta}_q)\|_2 \le r_{q,ij}, \\ 0, & \text{otherwise.} \end{cases} \tag{32}$$

The channel mask is the union of all dots. Alternative dot shapes replace the Euclidean norm with a shape-specific boundary or a scaled vector primitive.

3.5.1 Screen angle

If the halftone grid is rotated by angle β, each centre is transformed around the image centre $\mathbf{c}$:

$$\mathbf{p}'_{ij} = \mathbf{c} + \mathbf{R}(\beta)(\mathbf{p}_{ij} - \mathbf{c}), \qquad \mathbf{R}(\beta) = \begin{bmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{bmatrix}. \tag{33}$$

Rotation changes the texture and reduces direct alignment with source or display axes. It also requires adequate overscan so that rotated rows cover the corners.

3.6 Hexagonal halftone

The hexagonal mode uses a triangular lattice. One convenient basis is

$$\mathbf{a}_1 = P\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{a}_2 = P\begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}. \tag{34}$$

Sample centres are

$$\mathbf{p}_{ij} = \mathbf{p}_0 + i\mathbf{a}_1 + j\mathbf{a}_2. \tag{35}$$

The nearest-neighbour distance is P, and the area of one lattice cell is

$$A_{\text{cell}} = \left|\det\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix}\right| = \frac{\sqrt{3}}{2}P^2. \tag{36}$$

Compared with a square grid at the same nearest-neighbour spacing, the triangular lattice provides more samples per unit area and has six symmetric neighbour directions. This produces a less axis-biased texture. Radius mapping, Colour separation and antialiasing follow the same principles as the square halftone.

3.7 Spiral stripes

Percepta's spiral renderer is based on a continuous Archimedean spiral. In polar coordinates around the selected centre (x_c, y_c),

$$r(\theta) = r_0 + \sigma b\theta, \tag{37}$$

where r_0 is the initial radius, $\sigma = +1$ or -1 controls winding direction, and b determines the distance between successive turns. Since

$$r(\theta + 2\pi) - r(\theta) = 2\pi b, \tag{38}$$

the requested turn spacing T gives

$$b = \frac{T}{2\pi}. \tag{39}$$

A point on the centreline is

$$\mathbf{s}(\theta) = \begin{bmatrix} x_c + r(\theta)\cos\theta \\ y_c + r(\theta)\sin\theta \end{bmatrix}. \tag{40}$$

The tangent and normal directions are obtained from

$$\mathbf{t}(\theta) = \frac{\mathrm{d}\mathbf{s}/\mathrm{d}\theta}{\|\mathrm{d}\mathbf{s}/\mathrm{d}\theta\|_2}, \qquad \mathbf{n}(\theta) = \begin{bmatrix} -t_y(\theta) \\ t_x(\theta) \end{bmatrix}. \tag{41}$$

For each channel, the source is sampled along the path and the local half-width $h_q(\theta)$ is derived from intensity. The two boundaries of the channel ribbon are

$$\mathbf{s}_{q,\pm}(\theta) = \mathbf{s}(\theta) + \boldsymbol{\delta}_q(\theta) \pm h_q(\theta)\mathbf{n}(\theta), \tag{42}$$

with a separation vector naturally defined along the normal:

$$\boldsymbol{\delta}_q(\theta) = d_q \mathbf{n}(\theta). \tag{43}$$

Path smoothing filters $h_q(\theta)$ before boundary construction. For a discrete sequence $h_q[k]$, a moving weighted average is

$$\widetilde{h}_q[k] = \frac{\sum_{m=-K}^{K} \omega_m h_q[k-m]}{\sum_{m=-K}^{K} \omega_m}, \tag{44}$$

where K is related to the smoothing setting. This prevents high-frequency source noise from producing unstable or self-intersecting edges.

The centre controls are expressed as percentages of output dimensions:

$$x_c = \frac{p_x}{100} W, \qquad y_c = \frac{p_y}{100} H. \tag{45}$$

Moving the centre changes both composition and sampling. A centred spiral gives radial symmetry; an off-centre origin creates sweeping arcs and can place the tightest turns over a chosen feature.

Practical note

Spiral stripes are most stable when the local ribbon width remains comfortably below the turn spacing. If Strength, minimum width and Colour separation together make neighbouring turns overlap excessively, highlights flatten and the spiral loses its readable structure.

4 Controls, defaults and adaptive scaling

4.1 Why density has pattern-specific units

A single generic "density" number would be ambiguous because the renderers are constructed differently. Percepta therefore displays the physical or geometric quantity that the user actually controls.

Table 1: Reference density defaults at an output size of 1600 px.

Pattern	Meaning of the density field	Default	Animation reference
Vertical stripes	Number of carriers across the output	14	$8 \to 24$
Horizontal stripes	Number of carriers across the output	14	$8 \to 24$
Diagonal stripes	Number of carriers across the projected diagonal span	18	$10 \to 28$
Halftone	Centre-to-centre dot spacing	12 px	24 px $\to$ 7 px
Hexagonal halftone	Nearest-neighbour dot spacing	12 px	24 px $\to$ 7 px
Spiral stripes	Distance between successive turns	19 px	24 px $\to$ 6 px

For stripes, a larger number means a denser pattern. For dots and spirals, a smaller spacing means a denser pattern. This reversal is intentional: the displayed value remains a directly interpretable geometric quantity.

4.2 Strength

Strength governs how strongly channel intensity changes the geometry. In the generic modulation of Equation (6), it scales the intensity-dependent term. Visually:

- low Strength produces similar widths or sizes over much of the image and weak tonal contrast;
- moderate Strength separates features while retaining coloured gaps and pattern identity;
- high Strength drives bright regions toward maximum coverage and dark regions toward minimum coverage.

Strength is not ordinary image contrast. Contrast modifies the source values before rendering; Strength changes how those values occupy geometric space. Their effects interact, so increasing both simultaneously can cause rapid saturation.

A useful diagnostic is the fraction of samples clipped to either limit. If a_q is the modulation amplitude, define

$$\rho_{\text{clip}} = \frac{1}{3WH} \sum_{q \in \{R,G,B\}} \sum_{x,y} \mathbf{1}\left[a_q(x,y) \in \{0,1\}\right], \tag{46}$$

where $\mathbf{1}[\cdot]$ is an indicator function. A very large ρ_{clip} means the output is using only extreme geometry and subtle source differences have been lost.

4.3 Colour separation

The separation setting is measured in output pixels. Its relative magnitude is therefore

$$d_{\text{rel}} = \frac{d}{L}, \tag{47}$$

where L is a characteristic output dimension. A fixed 10 px separation has twice the relative effect on an 800 px image as on a 1600 px image. Adaptive scaling can preserve appearance by scaling d with output size, although a user-edited value should generally remain under explicit control.

Small separation exposes coloured edges while maintaining substantial local overlap. Large separation creates a pronounced chromatic offset but increases the size of the neighbourhood required for visual averaging. It may also reveal three independent copies of high-contrast features.

Practical note

When a result is attractive close up but fails to reconstruct, first reduce Colour separation before changing the pattern. This often restores local RGB balance without removing the chosen geometric character.

4.4 Source contrast

Contrast expands or compresses source values around the mid-point. It is useful when a source is flat or when the geometric renderer suppresses weak differences. It should be adjusted while watching both dark and bright areas.

The contrast gain of 1.55 visible in the reference captures is intentionally assertive. It helps the example portrait survive geometric sampling, but a source already containing deep blacks and bright highlights may require a lower value.

A histogram-based interpretation is helpful. If the input channel has probability density $p_C(c)$, the linear contrast transform changes the occupied range. Clamping accumulates probability at zero and one, producing flat black gaps or maximum-width geometry. The optimal value is therefore source-dependent.

4.5 Output size

Output size specifies the raster dimensions used for preview generation and export. For a square crop, $W = H = L$. Other crop modes use the selected aspect ratio while preserving the reference dimension convention used by the application.

Increasing size provides:

- more pixels along each edge and smoother raster geometry;
- the ability to print larger at a given pixel density;
- more accurate representation of thin RGB fringes;
- higher memory use and longer rendering/export time.

If the output contains WH pixels, a basic 8-bit RGB array requires approximately

$$M_8 = 3WH \text{ bytes.} \tag{48}$$

A floating-point working array with three 32-bit channels requires

$$M_{32} = 12WH \text{ bytes,} \tag{49}$$

not including source images, masks, antialiasing buffers or animation frames. At 4000×4000, one RGB float buffer alone occupies about 192 MB.

4.6 Adaptive defaults

The agreed reference size is

$$L_0 = 1600 \text{ px.} \tag{50}$$

To preserve approximately the same apparent geometry when the output dimension changes to L, quantities expressed as *counts* scale upward with size, whereas quantities expressed as *pixel spacing* also scale in pixels so that their normalised spacing remains constant.

For a reference stripe count N_0, an adaptive count can be defined as

$$N(L) = \max\left(1, \text{round}\left[N_0 \frac{L}{L_0}\right]\right). \tag{51}$$

For a reference spacing P_0,

$$P(L) = P_0 \frac{L}{L_0}. \tag{52}$$

The same scaling applies to default RGB separation and minimum feature dimensions when the goal is scale invariance.

For example, a 12 px dot spacing at 1600 px corresponds to

$$P(3200) = 12 \times \frac{3200}{1600} = 24 \text{ px.} \tag{53}$$

Although the pixel spacing doubles, the output also doubles, so the number of dots across the image remains comparable.

4.6.1 Preserving user intent

Automatic defaults should update when the user changes output size only while the relevant field remains in an automatic state. Once the user explicitly edits a field, silently overwriting it would be surprising. A robust state model uses an “automatic/manual” flag for each adaptive parameter:

$$P_{\text{displayed}} = \begin{cases} P_0 L / L_0, & \text{automatic state,} \\ P_{\text{user}}, & \text{manual state.} \end{cases} \tag{54}$$

Resetting the pattern can restore automatic defaults.

4.7 Framing controls

Table 2: Framing controls and their mathematical role.

Control	Operation	Typical use
Crop	Selects output aspect ratio and visible source window	Square artwork, portrait, landscape or custom framing
Zoom	Isotropic scale z in Equation (1)	Make the subject fill the geometric bandwidth
Pan X/Y	Translation before source resampling	Place eyes, face or focal object relative to carriers
Rotation	Rotation φ around the crop centre	Straighten the image or avoid alignment with carriers
Reset	Restores neutral framing	Return to a reproducible starting point

Crop and framing are not cosmetic post-processing. They determine which source information is sampled by the pattern. A small, centred subject generally survives geometric encoding better than one occupying a minor fraction of a busy frame.

4.8 Pattern-specific controls

4.8.1 Line smoothing

Smoothing acts along the carrier direction and suppresses abrupt changes in width. If raw widths are $w[k]$, a simple exponential smoother is

$$\widetilde{w}[k] = \lambda w[k] + (1-\lambda)\widetilde{w}[k-1], \qquad 0 < \lambda \le 1. \tag{55}$$

Small λ gives stronger smoothing. Excessive smoothing removes fine detail and can shift narrow features.

4.8.2 Halftone shape, angle and minimum size

Dot shape changes the support function used in Equation (32). Circular dots are rotationally neutral; square or polygonal shapes create stronger directional texture. Screen angle rotates the lattice. Minimum size ensures weak but nonzero channel values remain visible, though it also raises the background level and reduces dark contrast.

4.8.3 Spiral centre, direction and path smoothing

The centre changes the polar coordinate origin, direction changes σ in Equation (37), and path smoothing filters local ribbon thickness as in Equation (44). Direction alone should not change the average reconstruction, but it changes the flow of visible fringes and can interact with asymmetric source structure.

5 Perceptual reconstruction

5.1 Two simultaneous spatial scales

The defining feature of a Percepta image is that it supports two readings. At high spatial resolution, the observer resolves individual RGB elements and sees the construction. At lower spatial resolution, the same elements are integrated and their local average forms the source image.

This is not a binary switch at a precise distance. Visibility changes continuously with pattern period, contrast, display or print size, viewing distance, ocular optics, retinal sampling and neural contrast sensitivity. The most interesting results often occupy an intermediate regime in which both pattern and portrait are apparent.

5.2 Low-pass approximation

A first-order model represents the combined optical and neural integration by a point-spread function $h_v(x, y)$. The perceived image is

$$\mathbf{I}_{\text{seen}}(x, y; D) = \left(h_v(\cdot, \cdot; D) * \mathbf{I}_{\text{out}}\right)(x, y), \tag{56}$$

where the dependence on viewing distance D indicates that one output pixel subtends a smaller visual angle as distance increases. A Gaussian approximation is

$$h_v(x, y; \sigma) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{x^2 + y^2}{2\sigma^2}\right]. \tag{57}$$

Its Fourier transform is also Gaussian:

$$H_v(f_x, f_y) = \exp\left[-2\pi^2\sigma^2(f_x^2 + f_y^2)\right], \tag{58}$$

which attenuates high spatial frequencies more strongly than low ones. The geometric carrier lies at relatively high frequency, while the slowly varying source envelope lies at lower frequency. Increasing distance therefore suppresses the visible carrier before it suppresses the reconstructed subject.

The visual system is not exactly Gaussian and contrast sensitivity is band-pass rather than a simple monotonic low-pass function. Equation (56) is nevertheless a useful explanatory and preview model.

5.3 Visual angle and cycles per degree

Let a physical pattern period be p millimetres and viewing distance be D millimetres. The angular period is

$$\theta_p = 2 \arctan\left(\frac{p}{2D}\right). \tag{59}$$

For small angles,

$$\theta_p \approx \frac{p}{D} \quad \text{radians.} \tag{60}$$

Converting to degrees, the corresponding spatial frequency in cycles per degree is approximately

$$f_{\text{cpd}} \approx \frac{\pi D}{180p}. \tag{61}$$

Thus, doubling viewing distance doubles the carrier frequency expressed in cycles per degree and makes it harder to resolve. For a digital display with pixel pitch p_{px} and a geometric period of P pixels,

$$p = P\,p_{\text{px}}, \tag{62}$$

so Equation (61) becomes

$$f_{\text{cpd}} \approx \frac{\pi D}{180 P p_{\text{px}}}. \tag{63}$$

For print, pixel pitch is replaced by the physical output dimension divided by the raster dimension.

5.4 Example viewing-distance calculation

Suppose a 1600 px square image is printed 240 mm wide. One output pixel corresponds to

$$p_{\text{px}} = \frac{240}{1600} = 0.15 \text{ mm.} \tag{64}$$

A 12 px halftone spacing corresponds to

$$p = 12 \times 0.15 = 1.8 \text{ mm.} \tag{65}$$

At $D = 1000$ mm,

$$f_{\text{cpd}} \approx \frac{\pi \times 1000}{180 \times 1.8} \approx 9.7 \text{ cycles/degree.} \tag{66}$$

The dots may still be visible to an observer with normal acuity, but their chromatic substructure and local size differences are substantially integrated. At two metres, the frequency doubles to approximately 19.4 cycles/degree and the pattern becomes much less individually resolvable.

This calculation is illustrative rather than prescriptive. Dot diameter, RGB separation, contrast, print quality and chromatic sensitivity all affect the actual transition.

5.5 Chromatic and luminance sensitivity

Human spatial resolution is generally higher for luminance variation than for purely chromatic red–green or blue–yellow variation. Narrow coloured fringes can therefore blend at distances where the broad light–dark envelope remains visible. This property benefits Percepta: the RGB carrier can lose its separate chromatic identity while the combined luminance and colour average continues to describe the source.

A standard linear approximation to relative luminance from linear RGB is

$$Y(x,y) = 0.2126R(x,y) + 0.7152G(x,y) + 0.0722B(x,y). \tag{67}$$

The coefficients reflect the strong photopic contribution of green and the weaker contribution of blue. Display RGB values are often gamma-encoded, so physically accurate luminance calculation requires linearisation first. If a channel value C_s is in standard sRGB form, the linear value C_ℓ is approximately

$$C_\ell = \begin{cases} \dfrac{C_s}{12.92}, & C_s \le 0.04045, \\ \left(\dfrac{C_s + 0.055}{1.055}\right)^{2.4}, & C_s > 0.04045. \end{cases} \tag{68}$$

Percepta's artistic rendering can operate directly on normalised channel values, but linear-light reasoning is useful when calibrating coverage so that area averages correspond more closely to emitted or reflected light.

5.6 Coverage, gamma and apparent brightness

If a display emits linear primary intensity over a fraction a of a local cell and black elsewhere, the physical mean channel intensity is proportional to a. On a gamma-encoded image, however, writing a value of one in covered pixels and zero elsewhere does not imply that all downstream systems preserve that linear mean. Display transfer functions, printer dot gain and image resampling can alter it.

A calibrated coverage function can therefore be written

$$a_q = F_q^{-1}(C'_q), \tag{69}$$

where F_q is the measured relationship between geometric coverage and perceived or printed channel response. The current application emphasises direct creative control rather than device-specific calibration, but Equation (69) identifies a possible future extension.

5.7 Why density changes recognisability

Let the carrier period be P pixels and a source feature have characteristic width L_f pixels. The number of pattern periods sampling that feature is roughly

$$n_f = \frac{L_f}{P}. \tag{70}$$

When $n_f < 1$, the feature can fall between samples or be represented by only part of one geometric element. When n_f is several units or more, its variation is distributed across multiple elements and is easier to reconstruct.

Increasing stripe count reduces P and increases n_f. Decreasing dot or turn spacing has the same effect. The trade-off is that fine carriers become less visible as artwork at close range and can become vulnerable to display or printer resolution limits.

5.8 Colour separation and integration neighbourhood

If red and blue structures are separated by a distance $2d$, a visual averaging kernel must span a comparable scale before their contributions recombine. In the Gaussian model, the relative attenuation of a carrier frequency f is given by Equation (58). Increasing separation introduces lower-frequency chromatic differences, which require stronger blur or greater viewing distance to suppress.

This explains why Colour separation and density should be adjusted together. A dense pattern with moderate separation can reconstruct at a convenient distance. A sparse pattern with large separation may require the observer to move far enough away that source detail is also lost.

5.9 A computational preview of distant viewing

A useful software preview is to downsample the output and then enlarge it for inspection. Let $\mathcal{D}_k$ be area downsampling by factor k and $\mathcal{U}_k$ be display enlargement. The preview is

$$\mathbf{I}_{\text{preview}} = \mathcal{U}_k\left(\mathcal{D}_k(\mathbf{I}_{\text{out}})\right). \tag{71}$$

Area downsampling approximates local integration. A Gaussian blur followed by downsampling better suppresses aliasing:

$$\mathbf{I}_{\text{preview}} = \mathcal{U}_k\left[\mathcal{D}_k\left(h_\sigma * \mathbf{I}_{\text{out}}\right)\right]. \tag{72}$$

This preview cannot reproduce an observer's exact contrast sensitivity or a print's physical behaviour, but it provides a rapid estimate of whether local RGB averages reconstruct the source.

Principle

The illusion is strongest when the geometric carrier is high enough in spatial frequency to be integrated at the intended viewing distance, while source features remain low enough in frequency to survive that integration.

6 Density animation and export

6.1 Animation purpose

A static Percepta image asks the observer to change viewing distance or zoom. Density animation performs a related transformation in time. It changes the spatial scale of the geometry while retaining the same source and rendering family. A sparse opening frame foregrounds the construction; a denser frame carries more source detail. The animation can therefore reveal how the illusion is built.

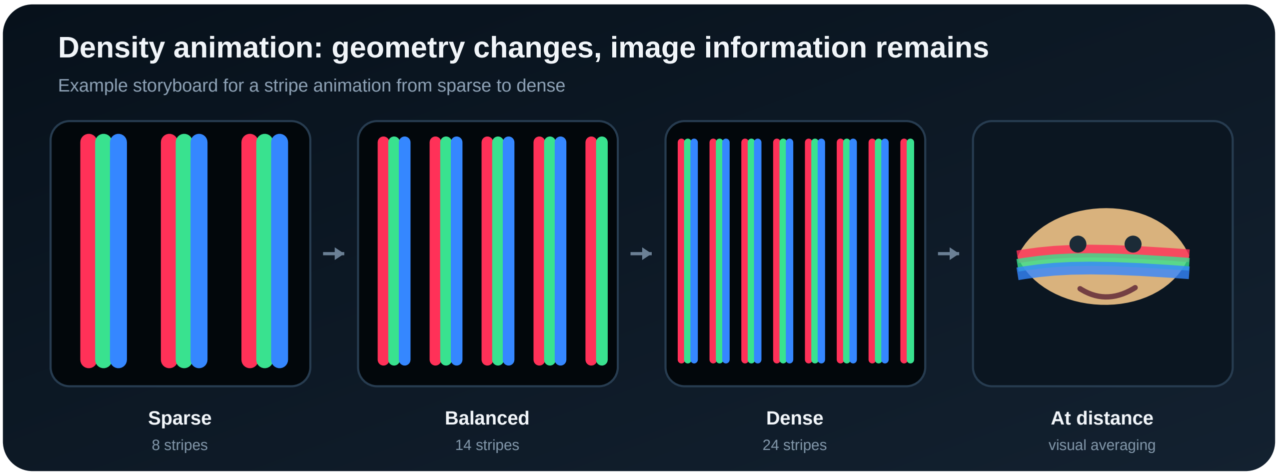

Figure 3: Storyboard of a stripe-density animation. The application interpolates the active density parameter, not the source image.

6.2 Time normalisation

Let the requested duration be T seconds and the time of a frame be $t \in [0, T]$. Normalised time is

$$u = \operatorname{clamp}\left(\frac{t}{T}, 0, 1\right). \tag{73}$$

An easing function $E(u)$ maps this interval to an interpolation phase. Linear interpolation uses

$$E_{\mathrm{lin}}(u) = u. \tag{74}$$

A smoothstep easing with zero first derivative at both ends is

$$E_{\mathrm{smooth}}(u) = 3u^2 - 2u^3. \tag{75}$$

A stronger quintic smoother is

$$E_{\mathrm{smoother}}(u) = 6u^5 - 15u^4 + 10u^3. \tag{76}$$

The user-facing easing selector can expose one or more of these choices. Linear motion makes parameter speed constant; smooth easing makes the start and end less abrupt.

6.3 Parameter interpolation

For start value p_0 and end value p_1,

$$p(t) = p_0 + (p_1 - p_0)\, E\left(\frac{t}{T}\right). \tag{77}$$

Stripe counts must be integer-valued, so the rendered count is

$$N(t) = \max(1, \mathrm{round}[p(t)]). \tag{78}$$

Dot and turn spacing can remain real-valued until geometry is rasterised. This produces smoother animation than rounding spacing to integer pixels at every frame.

For halftone and spiral examples, animation commonly proceeds from a large spacing to a small spacing. Although the numeric value decreases, geometric density increases.

6.4 Ping-pong phase

Ping-pong playback traverses the range forward and then backward. For a repeating phase $u \in [0, 1)$, a triangular phase is

$$u_{\triangle} = 1 - |2u - 1|. \tag{79}$$

The animated parameter becomes

$$p_{\mathrm{pp}}(t) = p_0 + (p_1 - p_0)\, E(u_{\triangle}). \tag{80}$$

If the exported duration T is intended to include a full forward-and-back cycle, u should span the triangle once over T. If T describes only the forward leg, the full ping-pong duration is $2T$. The interface and export metadata should use one convention consistently.

6.5 Frame rate and frame count

At frame rate F frames per second, a practical inclusive frame count is

$$N_f = \lfloor TF \rfloor + 1. \tag{81}$$

Frame times can be defined by

$$t_k = \frac{k}{N_f - 1} T, \qquad k = 0, 1, \ldots, N_f - 1. \tag{82}$$

This includes the exact start and end states. For seamless looping, duplicating the terminal frame may cause a visible pause. Loop export can omit one duplicate endpoint or use timing metadata to maintain constant motion.

The raw uncompressed memory required to hold all RGB frames is approximately

$$M_{\mathrm{anim}} = 3WHN_f \text{ bytes} \tag{83}$$

for 8-bit RGB. A 1600 px square, four-second animation at 12 FPS contains about 49 frames and would require approximately 376 MB if every frame were retained uncompressed. Streaming frames directly to the encoder is therefore preferable.

6.6 Recommended animation ranges

The following ranges are designed to show a meaningful transition without passing through excessively sparse or computationally wasteful states.

Table 3: Reference density animation ranges at 1600 px.

Pattern	Start	End	Interpretation
Vertical stripes	8	24	Threefold increase in carrier count
Horizontal stripes	8	24	Threefold increase in carrier count
Diagonal stripes	10	28	Denser range compensates for projected span
Halftone	24 px	7 px	Coarse dots become a detailed screen
Hexagonal halftone	24 px	7 px	Coarse triangular lattice becomes dense
Spiral stripes	24 px	6 px	Widely spaced turns tighten around the image

At higher output sizes, these values can be adaptively scaled using Equations (51) and (52).

6.7 Preview, playback and saving

The animation controls serve distinct purposes:

- **Preview** prepares or displays a reduced-resolution sequence for rapid evaluation.
- **Play** advances frames from the current position.
- **Pause** retains the current frame and allows playback to resume.
- **Stop** returns to a defined initial state.
- **Save** renders the requested output resolution and passes frames to the selected encoder.

Preview should prioritise responsiveness; final save should prioritise geometric and colour fidelity. A reduced preview can use lower output size or fewer frames, provided the interface clearly indicates that final export is rendered separately.

6.8 Still-image export formats

6.8.1 PNG

PNG is a lossless raster format suitable for the web, presentations and general image workflows. It preserves narrow coloured structures without JPEG blocking or chroma subsampling. File size can be large for detailed RGB patterns because the geometry contains many sharp colour transitions.

6.8.2 TIFF

TIFF is appropriate for high-resolution masters and print workflows. Depending on the writer, it can store lossless compression, metadata and higher bit depth. A print provider may prefer TIFF when the image is ultimately rasterised.

6.8.3 SVG

SVG represents geometric primitives as vectors. It is valuable for stripes, circles and spiral paths because the result can be scaled without pixelation. However, file size and rendering complexity can become high when every small dot or path segment is represented separately. Effects implemented only in raster form must either be approximated or embedded as raster content.

6.8.4 PDF

PDF is convenient for page-based distribution and professional printing. Vector-capable renderers can retain scalable geometry; raster output can also be embedded at a controlled physical size. The PDF page and artwork dimensions should be set intentionally rather than relying on a viewer's default scaling.

6.9 Physical print size and raster resolution

If the raster width is W pixels and the print width is S inches, the effective pixel density is

$$\mathrm{PPI} = \frac{W}{S}. \tag{84}$$

Conversely,

$$S = \frac{W}{\mathrm{PPI}}. \tag{85}$$

A 4000 px output printed at 300 PPI is approximately 13.3 inches (33.9 cm) wide. At 200 PPI it is 20 inches (50.8 cm) wide. Geometric art can often tolerate lower PPI than photography, but thin RGB gaps must remain resolvable by the printer.

The physical period of a P-pixel pattern element is

$$p_{\mathrm{mm}} = 25.4\frac{P}{\mathrm{PPI}}. \tag{86}$$

This quantity, together with viewing distance, determines visual angle through Equation (59).

6.10 Colour management and printing

Percepta is based on additive RGB mixing on screen. Printing uses subtractive inks and introduces dot gain, paper colour, ink limits and a device profile. A printed result can therefore differ from its emissive display preview.

Recommended practice:

1. retain an RGB master and avoid repeated format conversions;
2. use a colour-managed application for proofing;
3. ask the printer for the intended ICC profile and rendering conditions;
4. make a small physical proof at final scale;
5. verify that the thinnest coloured structures do not disappear or merge;
6. disable uncontrolled "photo enhancement" in consumer printer drivers.

Limitation

An SVG or PDF is not automatically print-safe. Vector geometry can be thinner than the output device can reproduce, and saturated RGB colours may lie outside the printer gamut. Physical proofing remains essential for important work.

7 Practical method, quality control and limitations

7.1 Choosing a source image

Percepta can process any conventional image, but not every source produces an equally convincing illusion. The encoder has finite spatial bandwidth: details smaller than the pattern period cannot be represented reliably. Sources with a clear hierarchy therefore perform best.

Favour images with:

- one dominant subject occupying a substantial fraction of the frame;
- recognisable silhouettes and medium-scale features;
- adequate separation between subject and background;
- controlled highlights and shadows;
- colour variation that remains meaningful after local averaging.

Potentially difficult sources include distant crowds, foliage, text-heavy scenes, repeating fabrics, fine hair against a busy background and images whose identity depends on one-pixel-scale detail.

7.2 Pattern selection by subject

Table 4: Starting pattern recommendations.

Subject	Recommended starting modes	Reason
Portrait	Vertical stripes, Halftone, Spiral	Faces tolerate controlled simplification and are recognised from medium-scale features
Landscape	Horizontal stripes, Hexagonal halftone	Horizons and broad tonal regions align naturally with lateral or isotropic sampling
Architecture	Diagonal stripes, Hexagonal halftone	Diagonal carriers avoid simple axis coincidence; hex grids reduce directional bias
Graphic logo	Vertical/Horizontal stripes, Halftone	Flat regions and strong boundaries translate clearly into coverage
Highly textured image	Dense halftone or dense stripes	More samples are required to preserve local variation
Abstract source	Spiral or strong Colour separation	Pattern character can take precedence over literal reconstruction

These are starting points rather than rules. A deliberate mismatch between subject and carrier can be aesthetically productive.

7.3 A controlled optimisation sequence

Changing several controls at once makes it difficult to identify the cause of improvement. The following one-variable sequence is recommended.

1. **Lock framing.** Place the subject and do not change the crop during the first comparison.
2. **Compare pattern families at their defaults.** Choose the structure whose close-range appearance suits the project.
3. **Sweep density.** Test at least one sparse, one reference and one dense value.
4. **Set Strength.** Increase until important tonal regions separate, then step back if highlights merge.
5. **Set Source contrast.** Use the smallest increase that restores recognisability.
6. **Add Colour separation.** Stop before the channels become independent copies at the intended viewing distance.
7. **Tune pattern-specific smoothing or minimum size.**
8. **Test final dimensions.** Increase output size and check adaptive values.
9. **Proof the export.** Inspect actual PNG/TIFF/SVG/PDF output, not only the live preview.

7.4 Quantitative quality indicators

Perceptual judgement remains central, but several numerical indicators can help diagnose a result.

7.4.1 Low-pass reconstruction error

Let $\mathcal{L}$ be a low-pass operator approximating distant viewing. Compare the filtered output with a filtered version of the framed source:

$$E_{\mathrm{LP}} = \frac{1}{3WH} \sum_{x,y} \left\| \mathcal{L}[\mathbf{I}_{\mathrm{out}}](x,y) - \mathcal{L}[\mathbf{C}'](x,y) \right\|_2^2 . \tag{87}$$

A smaller value indicates better preservation at the selected integration scale. It does not measure the artistic quality of the visible pattern.

7.4.2 Structural carrier visibility

A high-pass operator $\mathcal{H}$ can estimate how strongly the geometric pattern remains visible:

$$V_{\mathrm{carrier}} = \frac{1}{3WH} \sum_{x,y} \| \mathcal{H}[\mathbf{I}_{\mathrm{out}}](x,y) \|_2^2 . \tag{88}$$

Percepta's artistic objective is not to minimise V_{carrier}. A successful result often balances low reconstruction error with enough high-frequency energy to make the construction visually compelling.

7.4.3 Gamut and clipping rate

The clipping fraction in Equation (46) can be extended to exported RGB values. Large areas at exact 0 or 1 may be intended, but unexpected clipping can reveal excessive contrast or overlapping channel geometry.

7.5 Troubleshooting guide

Symptom	Likely cause	Corrective action
Source is not recognisable	Pattern too sparse; source too small in frame; weak tonal separation	Increase stripe count or reduce spacing; crop tighter; raise contrast moderately
Image becomes nearly white	Excessive Strength, contrast or overlap	Lower Strength; reduce contrast; reduce minimum size; inspect bright-source clipping
RGB copies are visible separately	Colour separation too large for density/viewing distance	Reduce separation; use denser geometry; increase viewing distance
Dark regions contain too much colour	Minimum geometry too large or smoothing spreads bright features	Reduce minimum size; reduce smoothing; inspect source black level
Thin lines flicker or break	Insufficient raster resolution or antialiasing	Increase output size; enable smoothing; export vector when appropriate
Diagonal edges look jagged	Raster grid aliasing	Increase supersampling/output size; use antialiased vector paths
Spiral turns collide	Turn spacing too small relative to width/separation	Increase turn spacing; lower Strength or separation
Animation pauses at loop point	Duplicate terminal frame or endpoint easing	Omit duplicated endpoint; adjust loop timing; use ping-pong phase consistently
Export is unexpectedly large	Dense vector geometry or many animation frames	Use lossless raster; reduce dimensions/FPS/duration; stream frames to encoder
Print loses coloured gaps	Printer resolution, dot gain or scaling	Make a final-size proof; increase feature width; avoid driver resampling

7.6 Computational complexity

For a raster output of $W \times H$ pixels, preprocessing and mask composition are generally $\mathcal{O}(WH)$. Stripe renderers can also be implemented in $\mathcal{O}(WH)$ by evaluating carrier phase per pixel or by rasterising paths whose total sampled length scales with image area at fixed density.

A halftone with spacing P contains approximately

$$N_{\text{dots}} \approx \frac{WH}{P^2} \tag{89}$$

for a square grid, and

$$N_{\text{hex}} \approx \frac{2WH}{\sqrt{3}P^2} \tag{90}$$

for a triangular lattice. Vector export cost grows with the number of primitives, whereas raster cost remains dominated by the output pixel count.

For an Archimedean spiral extending to radius $R_{\max}$ with turn spacing T, the number of turns is approximately

$$N_{\text{turns}} \approx \frac{R_{\max}}{T}. \tag{91}$$

The arc length from $\theta = 0$ to $\theta = \Theta$ is

$$\ell(\Theta) = \int_0^{\Theta} \sqrt{r(\theta)^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2}\,\mathrm{d}\theta. \tag{92}$$

Dense spirals therefore require many path samples and can be more expensive than simple stripes.

7.7 Known limitations

Limitation

Percepta is an artistic perceptual encoder, not a reversible compression method. Fine detail is intentionally discarded, and the original image cannot be recovered exactly from the output.

The principal limitations are:

- **Source dependence.** Optimal settings vary with composition, contrast and scale.
- **Observer dependence.** Visual acuity, colour vision and viewing conditions alter the effect.
- **Device dependence.** Displays and printers differ in gamut, resolution, transfer function and scaling.
- **Sampling loss.** Features smaller than the carrier period cannot be represented faithfully.
- **Edge interaction.** RGB separation can create coloured ghosts around high-contrast boundaries.
- **Raster/vector differences.** Antialiasing and path tessellation can cause small differences between preview and export.
- **Animation cost.** High-resolution sequences multiply rendering and encoding requirements.

7.8 Implementation and distribution

The current application is designed around a compact Python/PyQt6 project. A clean distribution can contain the executable source entry point and an assets folder, with the launcher responsible for component checks and packaged execution.

The runtime requirements used by the project include Python 3.10 or newer, PyQt6, NumPy, Pillow, imageio and imageio-ffmpeg. A packaged Windows build can use PyInstaller. The startup sequence may display the Percepta splash screen, check components once per application version and store state in the user's application-data directory.

Still-image export should be deterministic for a fixed source, framing state, pattern and parameter set. Reproducibility is improved by storing:

- application version;
- source file identity and dimensions;
- crop and affine framing parameters;
- pattern name and every common/pattern-specific parameter;
- output dimensions and export format;
- animation timing and easing when applicable.

A future preset or sidecar format could serialise these values in JSON.

7.9 Possible extensions

The existing model naturally supports several extensions without changing its core identity:

- perceptual or device-calibrated coverage functions F_q^{-1};
- automatic density selection based on source spatial frequency;
- GPU rasterisation for high-resolution animation;
- batch rendering and preset comparison sheets;
- objective low-pass reconstruction previews and error maps;
- additional continuous paths or lattice families;
- colour-management metadata and printer-specific proof modes.

These are development possibilities rather than current user features.

8 Glossary, notation and references

8.1 Compact glossary

Additive RGB composition. Combination of red, green and blue light; full coincident channels produce white on an emissive display.

Antialiasing. Partial edge coverage or supersampling used to reduce stair-stepping and flicker.

Carrier. Repeated line, lattice or path whose local width or size carries image information.

Chromatic fringe. Coloured boundary produced by relative RGB displacement.

Coverage. Fraction of a local cell occupied by one channel's geometry.

Cycles per degree. Number of repeated cycles subtending one degree of visual angle.

Density. Stripe count, dot spacing or spiral turn spacing, depending on the renderer.

Duty cycle. Fraction of a period occupied by a stripe.

Halftone. Continuous-tone representation using variable-size repeated dots.

Point-spread function. Spatial response to an ideal point, used here as an approximation of optical and neural blur.

Spatial frequency. Rate of repetition or change across visual space.

Strength. Amplitude of intensity-to-geometry modulation.

8.2 Further reading

References

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8.3 Principal notation

Symbol	Meaning	Symbol	Meaning	Symbol	Meaning
W, H	Output dimensions	x, y	Output coordinates	u, v	Source coordinates
R, G, B	Source channels	M_q	Channel mask	a_q	Geometric amplitude
P	Carrier spacing	N	Stripe count	d	Colour separation
r	Dot/polar radius	T	Turn spacing or duration	F	Frames per second
D	Viewing distance	f_{cpd}	Cycles per degree	L_0	1600 px reference size

8.4 Document status

This manual describes Percepta version 0.31.0 and the interface supplied in July 2026. The equations form a coherent technical specification of observed behaviour; they are not a verbatim listing of source-code operations. Default values and figures should be updated when the application changes.

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